

Upstream Production Performance Analysis

*Production loss rarely
happens suddenly.*

The warning signs are already in the data

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Tool Used : Microsoft Power BI

22

Wells

8.2Mbbl

Oil Produced

95%

Average Uptime%

9,420

Avg Daily Production

Dataset Overview

01

Dataset Type

Upstream Production Dataset

02

Total Rows

1,391

03

Total Columns: 26

Basin, Operator, Field Name, Formation, Spud Date, Report Date, API-Gravity, etc

04

Key Metrics

Oil Production, Gas Production, Uptime%, Downtime Hours, Lifting Cost, Water-Cut, etc

05

Data Range

2021 - 2024 Operational Data

Project Overview

This Project analyzes oil production operations across multiple wells and fields to uncover trends in production performance, operational efficiency, downtime behavior, and well reliability.

The analysis focuses on Identifying operational risk, evaluating well performance, and transforming raw operational data into actionable business insights.

Key Business Questions

- 01 Which fields are producing most efficiently?
- 02 Do older wells experience higher downtime?
- 03 Which wells show signs of operational risk?
- 04 How does downtime impact operational performance?
- 05 Are high-producing wells also cost-efficient?

Data Preparation & Modeling

The raw operational dataset was cleaned, transformed, and structured into a star-schema model to support efficient reporting and analytical calculations.

Cleaned Data

Well_ID	Field_Name	Basin	Operator	Well_Type	Formation	Fluid_Type
1	1 Agbami	Deepwater OML	Chevron	Producer	Agbada	Crude Oil
2	1 Agbami	Deepwater OML	Chevron	Producer	Agbada	Crude Oil
3	1 Agbami	Deepwater OML	Chevron	Producer	Agbada	Crude Oil
4	1 Agbami	Deepwater OML	Chevron	Producer	Agbada	Crude Oil
5	1 Agbami	Deepwater OML	Chevron	Producer	Agbada	Crude Oil
6	2 Agbami	Deepwater OML	Chevron	Producer	Afam	Gas Cond
7	2 Agbami	Deepwater OML	Chevron	Producer	Afam	Gas Cond
8	2 Agbami	Deepwater OML	Chevron	Producer	Afam	Gas Cond
9	2 Agbami	Deepwater OML	Chevron	Producer	Afam	Gas Cond
10	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil
11	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil
12	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil
13	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil
14	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil
15	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil
16	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil
17	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil
18	3 Agbami	Deepwater OML	Chevron	Producer	Benin	Crude Oil

Few KPI/ Measures

Data

Search

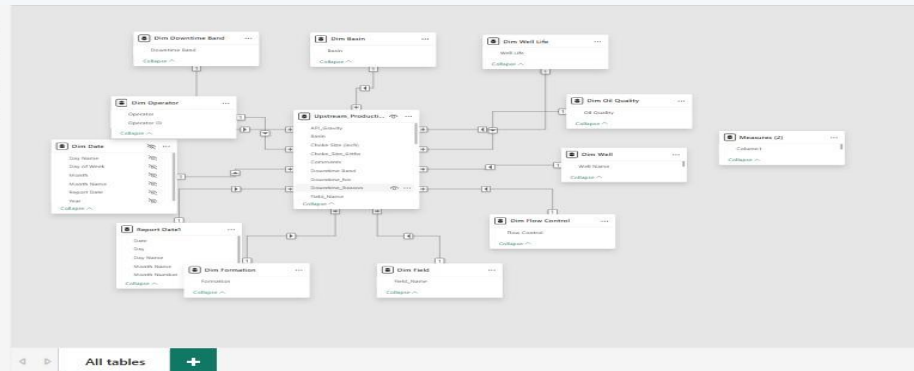
- ☐ uptime py
- ☐ Uptime_hrs
- ☐ yoy tDT
- ☐ Column1
- ☐ Measure
- ☐ Py oil.p
- ☒ total measure
 - ☐ Downtime Incidents
 - ☐ Total Downtime
 - ☐ Total Gas Production
 - ☐ total lifting
 - ☐ Total Oil Production
 - ☐ Total Water Production
 - ☐ year dis
 - ☐ year dis color
 - ☐ YOY oil.p
 - ☐ YOY oil.p2
- ☒ Report Date1
- ☒ Upstream_Production

Data Preparation Process

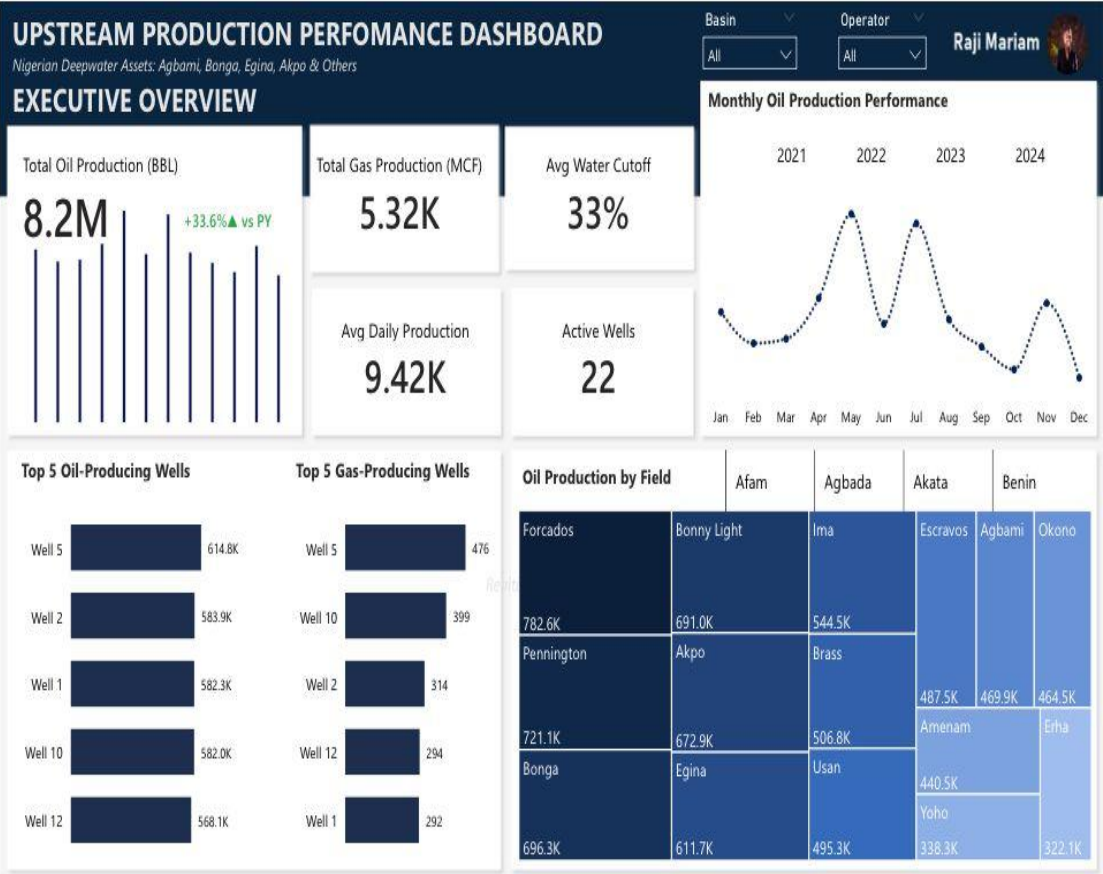
- Cleaned and transformed raw operational data
- Created fact and dimension tables
- Built relationships for filtering and analysis
- Developed DAX measures for KPIs and operational insights

Null Values in the Dataset were filled up with median of that particular column

Data Modelling:- Star- Shema



Executive Overview



Operational Efficiency



Key Highlight

Overall Average Uptime is impressive at **95%**, indicating good operational stability.

Summary

However, **12,667** hours of total downtime were recorded, averaging 9 hours per event.

Clear pattern observed: Older wells (Mature wells) experience significantly higher downtime and lifting costs.

Major downtime causes include Hydrate Blockage, ESP Failure, Weather Downtime, and Sand Production.

Risk & Reliability

RISK & RELIABILITY

Well Risk & Reliability Overview

Operator

All

Field_Name

All

Year

All

Well Name		Oil Produced		Gas Produced		Uptime %		Lifting Cost \$		Average Water Cut%		Well Status
Well 9	↑	514.4K	↑	369.7		93%		\$1,529.0	↓	31%		Healthy
Well 8	↑	554.5K	↑	328.3		96%		\$1,690.8	↓	30%		Healthy
Well 7	↑	532.9K	↑	383.7		96%		\$1,704.2	↓	34%		Healthy
Well 6	↑	432.6K	→	264.5		97%		\$1,270.5	↓	32%		Healthy
Well 5	↑	614.8K	↑	475.8		92%		\$1,804.3	↓	37%		Healthy
Well 4	→	412.9K	→	317.0		95%		\$1,476.2	↓	39%		Healthy
Well 3	↑	490.6K	↑	407.1		97%		\$1,585.0	↓	39%		Healthy
Well 22	↓	18.7K	↓	32.0		98%		\$124.4	↑	50%		At Risk
Well 21	↓	24.5K	↓	28.2		99%		\$70.1	↑	46%		At Risk
Well 20	↓	76.0K	↓	63.0		99%		\$254.4	↓	35%		Healthy
Well 2	↑	583.9K	→	313.7		96%		\$1,517.9	↓	34%		Healthy
Well 19	↓	115.9K	↓	44.3		94%		\$308.4	↓	35%		Healthy
Well 18	↓	113.1K	↓	40.9		87%		\$293.8	↓	28%		Healthy
Well 17	↓	153.1K	↓	130.8		91%		\$479.8	↓	29%		Healthy
Well 16	→	230.9K	↓	165.8		96%		\$655.0	↓	32%		Healthy
Well 15	→	306.9K	→	197.6		97%		\$922.3	↓	28%		Healthy
Well 14	→	365.7K	→	210.4		95%		\$1,050.8	↓	37%		Healthy
Well 13	→	405.4K	→	249.9		96%		\$1,007.3	↓	29%		Healthy
Well 12	↑	568.1K	→	294.3		95%		\$1,291.8	↓	30%		Healthy
Well 11	↑	566.0K	→	307.6		91%		\$1,514.4	↓	30%		Healthy
Well 10	↑	582.0K	↑	399.2		95%		\$1,528.3	↓	27%		Healthy
Well 1	↑	582.3K	→	291.7		97%		\$1,499.3	↓	31%		Healthy

Key Highlight

This dashboard enables proactive risk identification, allowing the team to focus intervention efforts on a small number of underperforming wells before they deteriorate further.

Summary

Early warning signals are visible through combined monitoring of Uptime, Water Cut, and Lifting Cost.

A few wells (e.g. Well 21, Well 22) are flagged as “At Risk”, characterized by lower production and higher water cut

Majority of wells are in Healthy status.

Key Insights & Recommendation

Insights

- 1 Well age is a critical factor affecting downtime and lifting costs
- 2 Strong YoY oil production growth (+33.6%)
- 3 Few wells contribute disproportionately to total production
- 4 Early risk identification is now possible through integrated monitoring

Recommendations

- 1 Prioritize maintenance on mature wells to reduce downtime
- 2 Focus optimization efforts on Top 5 wells and leading fields
- 3 Implement regular reviews of “At Risk” wells
- 4 Consider predictive analytics for future downtime prevention
- 5 Monitor wells with increasing water cut